

STA 2023 - Final Exam Practice: Answer Key

Spring 2026

Question 1

- (a)
 - i. **Population:** All UCF students
 - ii. **Sample:** The 200 UCF students who volunteered
 - iii. **Parameter:** $\mu = 65$ (The true average stress level of all UCF students)
 - iv. **Statistic:** $\bar{x}_{before} = 66$ and $\bar{x}_{after} = 54$ (The average stress levels of the sample)
- (b)
 - i. Quantitative, Discrete
 - ii. Qualitative
 - iii. Quantitative, Continuous
 - iv. Qualitative
- (c)
 - i. Interval
 - ii. Nominal
 - iii. Ratio
- (d) **Designed experiment.** The researcher is actively applying a treatment (the wellness program) to the subjects to measure its effect on their stress levels, rather than just observing them without interference.
- (e) **No.** Because the participants were volunteers (not randomly assigned to a treatment and control group), there may be confounding variables (such as motivation to reduce stress) that caused the reduction. Without randomization and a control group, causation cannot be definitively established.
- (f)
 - i. Simple random sampling
 - ii. It removes selection bias and ensures the sample is more representative of the entire population, unlike a volunteer sample which is prone to volunteer bias.

Question 2

(a) **Five-number summary:**

- i. Minimum = 45
- ii. $Q_1 = 61.5$
- iii. Median $Q_2 = 72$
- iv. $Q_3 = 87.5$
- v. Maximum = 120

(b) **Boxplot Construction:**

Interquartile Range
$IQR = Q_3 - Q_1$
Outlier Fences
Lower Fence = $Q_1 - 1.5 \times IQR$ Upper Fence = $Q_3 + 1.5 \times IQR$

Substitution and Calculation:

- $IQR = 87.5 - 61.5 = 26$
- Lower Fence = $61.5 - 1.5(26) = 22.5$
- Upper Fence = $87.5 + 1.5(26) = 126.5$

Final Answer: A boxplot drawn with a box extending from $Q_1 = 61.5$ to $Q_3 = 87.5$, with a vertical line inside at the Median = 72. The left whisker extends to the Minimum = 45, and the right whisker extends to the Maximum = 120. Outliers: None.

(c) **Shape of the Distribution: Right-skewed.** The distance from the median to Q_3 ($87.5 - 72 = 15.5$) is greater than the distance from Q_1 to the median ($72 - 61.5 = 10.5$), and the right whisker ($120 - 72 = 48$) is longer than the left whisker ($72 - 45 = 27$), indicating a longer tail on the right side.

Question 3**Probability of Event A**

$$P(A) = \frac{n(A)}{N}$$

(a)

$$\text{i. } P(L) = \frac{120}{200} = 0.6000$$

$$\text{ii. } P(T) = \frac{80}{200} = 0.4000$$

$$\text{iii. } P(L \cap T) = \frac{50}{200} = 0.2500$$

General Addition Rule

$$P(L \cup T) = P(L) + P(T) - P(L \cap T)$$

(b)

$$P(L \cup T) = 0.6000 + 0.4000 - 0.2500 = 0.7500$$

Conditional Probability

$$P(T|L) = \frac{P(T \cap L)}{P(L)}$$

(c)

$$P(T|L) = \frac{0.2500}{0.6000} \approx 0.4167$$

Independence Check

$$P(L) \times P(T) = P(L \cap T)$$

(d)

Calculation: $0.6000 \times 0.4000 = 0.2400$.

No, the events are not independent because $0.2400 \neq 0.2500$.

Multiplication Rule (Without Replacement)

$$P(L_1 \cap L_2 \cap L_3) = P(L_1) \times P(L_2|L_1) \times P(L_3|L_1 \cap L_2)$$

(e)

$$P(L_1 \cap L_2 \cap L_3) = \left(\frac{120}{200}\right) \times \left(\frac{119}{199}\right) \times \left(\frac{118}{198}\right) \approx 0.2138$$

Question 4(a) **Probability Distribution of X :**

- $x = -2, P(X = -2) = \frac{2}{6} = 0.3333$
- $x = 1, P(X = 1) = \frac{3}{6} = 0.5000$
- $x = 5, P(X = 5) = \frac{1}{6} = 0.1667$

Expected Value

$$E(X) = \sum [x \times P(X = x)]$$

(b)

$$E(X) = (-2)(0.3333) + (1)(0.5000) + (5)(0.1667) \approx 0.6667$$

Standard Deviation of a Discrete Random Variable

$$\sigma = \sqrt{\sum [x^2 \times P(x)] - [E(X)]^2}$$

(c)

$$E(X^2) = (-2)^2(1/3) + (1)^2(1/2) + (5)^2(1/6) = 4/3 + 1/2 + 25/6 = 8/6 + 3/6 + 25/6 = 36/6 = 6$$

$$\sigma = \sqrt{6 - (2/3)^2} = \sqrt{6 - 4/9} = \sqrt{50/9} \approx 2.3570$$

Binomial Probability

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y}$$

(d)

$$n = 10, p = 0.5, P(Y \geq 3) = 1 - P(Y \leq 2)$$

$$P(Y \leq 2) = P(Y = 0) + P(Y = 1) + P(Y = 2) = 0.0009765 + 0.0097656 + 0.0439453 = 0.0546875$$

$$P(Y \geq 3) = 1 - 0.0546875 \approx 0.9453$$

Binomial Mean and Standard Deviation

$$\mu_Y = n \times p$$

$$\sigma_Y = \sqrt{n \times p \times (1 - p)}$$

(e)

$$\mu_Y = 10 \times 0.5 = 5$$

$$\sigma_Y = \sqrt{10 \times 0.5 \times 0.5} = \sqrt{2.5} \approx 1.5811$$

Question 5

Sampling Distribution Mean and Standard Error

$$\begin{aligned}\mu_{\bar{x}} &= \mu \\ \sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{n}}\end{aligned}$$

(a)

$$\mu_{\bar{x}} = 180$$

$$\sigma_{\bar{x}} = \frac{20}{\sqrt{64}} = 2.5$$

Z-score for Sample Mean

$$Z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

(b)

$$Z_1 = \frac{175 - 180}{2.5} = -2.00$$

$$Z_2 = \frac{185 - 180}{2.5} = 2.00$$

$$P(-2.00 < Z < 2.00) = 0.9772 - 0.0228 = 0.9544$$

(c) Normal Approximation Check:

Success-Failure Condition

$$n \times p \geq 10 \text{ and } n \times (1 - p) \geq 10$$

$$100 \times 0.30 = 30 \geq 10$$

$$100 \times 0.70 = 70 \geq 10$$

Yes, the condition is satisfied.

Sampling Distribution for Proportion

$$\begin{aligned}\mu_{\hat{p}} &= p \\ \sigma_{\hat{p}} &= \sqrt{\frac{p(1-p)}{n}}\end{aligned}$$

(d)

$$\mu_{\hat{p}} = 0.3000$$

$$\sigma_{\hat{p}} = \sqrt{\frac{0.30(0.70)}{100}} \approx 0.0458$$

Z-score for Sample Proportion

$$Z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}}$$

(e)

$$Z = \frac{0.35 - 0.30}{0.0458257} \approx 1.09$$

$$P(Z > 1.09) = 1 - 0.8621 = 0.1379$$

Question 6**Confidence Interval for Proportion**

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

(a)

Given: $n = 250, x = 165 \Rightarrow \hat{p} = 0.66, z_{0.025} = 1.96$

$$\text{Margin of Error (E)} = 1.96 \sqrt{\frac{0.66(0.34)}{250}} \approx 1.96(0.02996) \approx 0.0587$$

$$\text{CI: } 0.66 \pm 0.0587 = (0.6013, 0.7187)$$

- (b) **Interpretation:** We are 95% confident that the true proportion of all students who support having flu vaccination clinics on campus every semester lies between 60.13% and 71.87%.